The Empirical Distribution Function

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Introduction

The empirical distribution function (ECDF) is the simplest possible non-parametric estimator of a probability distribution: it puts mass $1/n$ at each observation and returns the resulting step-function CDF. Despite its simplicity, the ECDF is consistent uniformly for the true CDF (Glivenko-Cantelli), has a tight universal error bound (Dvoretzky-Kiefer-Wolfowitz), and underlies every non-parametric statistic built on ranks or percentiles.

Prerequisites

The reader should know what a CDF is and should be able to sort a vector in R.

Theory

For iid observations $X_1, \dots, X_n$ from a CDF F, the ECDF is

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{X_i \le x\}.$$

Properties at each fixed x:

- $n\hat{F}_n(x) \sim \text{Binomial}(n, F(x))$, so $E[\hat{F}_n(x)] = F(x)$ and $\text{Var}[\hat{F}_n(x)] = F(x)(1 - F(x))/n$.
- By the CLT, $\sqrt{n}(\hat{F}_n(x) - F(x)) \xrightarrow{d} \mathcal{N}(0, F(x)(1 - F(x)))$.

Dvoretzky-Kiefer-Wolfowitz (DKW) inequality. For every n and every $\varepsilon > 0$,

$$P\left(\sup_x |\hat{F}_n(x) - F(x)| > \varepsilon\right) \le 2e^{-2n\varepsilon^2}.$$

This is a *uniform* bound – it controls the supremum over all x – and holds for every F. It underwrites non-parametric confidence *bands* for the entire CDF, not just pointwise.

Sup-norm CI. Setting the DKW bound equal to α and solving for ε gives a $1 - \alpha$ confidence band of the form $[\hat{F}_n(x) - \varepsilon, \hat{F}_n(x) + \varepsilon]$ for all x simultaneously.

Assumptions

The ECDF is defined for any distribution; it is an estimator with minimal assumptions – only iid sampling (or exchangeability). Dependent data complicate the variance but not the point estimator.

R Implementation

```r
library(ggplot2)

set.seed(2026)
n <- 100
```

```r
alpha <- 0.05
x <- rnorm(n)

Fn <- ecdf(x)
eps <- sqrt(log(2 / alpha) / (2 * n))

grid <- seq(-4, 4, length.out = 1001)
df <- data.frame(
  x      = grid,
  Fhat   = Fn(grid),
  lower  = pmax(0, Fn(grid) - eps),
  upper  = pmin(1, Fn(grid) + eps),
  Ftrue  = pnorm(grid)
)

ggplot(df, aes(x = x)) +
  geom_step(aes(y = Fhat), colour = "steelblue") +
  geom_ribbon(aes(ymin = lower, ymax = upper), alpha = 0.2,
              fill = "steelblue") +
  geom_line(aes(y = Ftrue), colour = "red", linetype = "dashed") +
  labs(y = "CDF",
       title = sprintf("ECDF with 95%% DKW band (n = %d)", n)) +
  theme_minimal()
```

The plot overlays the ECDF, the DKW 95% band, and the true CDF. With $n = 100$ and $\alpha = 0.05$, the band width is $\varepsilon = \sqrt{\log(40)/200} \approx 0.136$ – constant across x.

Output & Results

The ECDF traces the true CDF closely; the DKW band contains it at every x, as expected since the DKW bound is universal. Increasing n to 1000 shrinks the band to $\varepsilon \approx 0.043$.

Interpretation

The ECDF is an omnibus, assumption-light summary of the data. Reporting a plot of the ECDF with DKW bands is a sensible way to communicate a distribution's shape when no parametric form is assumed. Quantile reporting, K-S tests, and many other non-parametric procedures ultimately derive from the ECDF.

Practical Tips

- For small samples ($n < 30$), the DKW band is wide; report pointwise Clopper-Pearson or Wilson CIs instead if you only need uncertainty at specific quantiles.
- The `ecdf()` function in R returns a function; apply it to any vector of x values to get empirical probabilities.
- Smooth variants (kernel-based) exist, but the ECDF's simplicity is a feature: it is the non-parametric MLE and a sufficient statistic in the non-parametric model.
- The DKW constant 2 is tight (Massart 1990); older texts quote larger constants.
- The ECDF is the foundation of quantile plots, Q-Q plots, and goodness-of-fit statistics like Kolmogorov-Smirnov and Cramer-von Mises.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**

- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr